Rajdeep (Ron) Sarma

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Education

PhD **Carnegie Mellon University**, Chemistry — Pittsburgh, USA, Aug 2022 – Present

- Principal Investigator: Dr Isaac Garcia-Bosch (GPA: 3.61/4.0) (IGB Lab)
- Topics: Physical Chemistry, Catalysis, Machine Learning

BS-MS **Indian Institute of Science Education and Research Bhopal**, Chemistry — Bhopal, India, Aug 2016 – May 2021

- Masters' Supervisor: Dr Debasish Manna (GPA: 7.85/10.0) (DM Lab)
- Topics: Chemical Biology, DNA (De)methylation, Inorganic Chemistry

Experience

Research Assistant, Team lead - Project EQUUS — Aug 2024 – Present

- Supervisors: Dr Isaac Garcia-Bosch and Dr Olexandr Isayev
- Lead a team of 7, developed a workflow for ultrafast theoretical screening of >800k organic molecules
- Tools used: [*Computational*] Python, PyTorch, RDKit, JavaScript, HTML, CSS, SQL, DFT (ωB97M-D3BJ/Def2-TZVPP), MLIP/NNP (AIMNet2). [*Experimental*] ^{1}H-NMR, Cyclic Voltammetry and Automated Experimentation

Research Assistant, Electron-Coupled Proton Buffers (ECPBs) — Jan 2023 – Aug 2024

- Supervisor: Dr Isaac Garcia-Bosch
- Experimental study of the thermochemistry of a new class of molecular systems: 6H+/6e– and 4H+/4e– ECPBs
- Tools used: [*Synthetic*] Organic and Inorganic Synthesis, Workups, Crystallizations, Glove-Box Techniques. [*Analytical*] ^{1}H-NMR, CV, Open-Circuit Potential, UV-vis Spectroscopy, Mössbauer and X-Ray Diffraction

Teaching Assistant — Aug 2022 - Dec 2023

- Course mentorship: 30 students (over 3 semesters). Research mentorship: 8 students (over varying durations)

Conferences

- NVIDIA GTC 2025: Attending and networking — Washington DC, Oct 2025
- RE+ 2025: NSF I-corps Regional funded visit for customer discovery — Las Vegas, Sept 2025
- ACS Fall 2025: Poster presentation on the updated Project EQUUS v1.1.0 — Washington DC, Aug 2025
- ACS Spring 2025: Two talks (6H+/6e– ECPBs and the initial Project EQUUS v1.0.0) — San Diego, March 2025

Publications

Stepwise 6H+/6e– Electron-Coupled-Proton Buffers based on Fe and Redox-Active Ligands
ACS Inorg. Chem., Sept 2025
Sarma, R., Wu, T., Ye, D., Qiu, Y. L., Park, S., Cohen, E., Xiong, J., Guo, Y., Garcia-Bosch, I.

Tuning the Thermochemistry and Reactivity of a Series of Cu-Based 4H+/4e– Electron-Coupled-Proton Buffers
ACS Inorg. Chem., 2024, doi: 10.1021/acs.inorgchem.4c00835
Wu, T., Puri, A., Qiu, Y. L., Ye, D., ***Sarma, R.***, Wang, Y., Kowalewski, T., Siegler, M., Stewart, M., Garica-Bosch, I.

A Chemical Model of a TET Enzyme for Selective Oxidation of Hydroxymethyl Cytosine to Formyl Cytosine
ACS Inorg. Chem., 2023, doi: 10.1021/acs.inorgchem.3c00063
Patil, D., Kundu, S., Pain, P. K., ***Sarma, R.***, Manna, D.

Courses

Introduction to Deep Learning - 11785 (Fall 25) Course Website

- Instructors: Dr Bhiskha Raj and Dr Rita Singh
- Tools Used: Python (OOP, NumPy, SciPy), PyTorch (data handling and preprocessing, optimizers, schedulers), Cloud Computing (AWS, GCP, Kaggle, Colab), WandB, Various Network Architectures (RNN, CNN, VAE, etc)
- Course Project: Dynamic Object Removal Using NeRF (*coming soon*)

Algorithms for Big Data - 15851 (Sring 25) Course Website

- Instructor: Dr David P Woodruff
- Tools Used: Linear Algebra, Multivariable Calculus, Probability Theory, Computational Complexity Theory, Sketching, Information Theory, Advanced ML techniques (Clustering based data-efficient training and HyperAttention)
- Course Project: Efficient Transformer-Based Equivariant Machine Learned Interatomic Potentials (Project Report)

Introduction to Machine Learning - 10701 (Fall 24) Course Website

- Instructors: Dr Geoffrey J Gordon and Dr Maria-Florin Balcan
- Tools Used: Python (OOP, NumPy, SciPy, Scikit-learn, PyTorch), Active Learning, Reinforcement Learning, Information Theory, VC dimensions, Rademacher Complexity, Clustering, Boosting, Various Model Architectures
- Course Project: Reinforcement Learning from Human Feedback (RLHF) with CartPole (Project Report)

Neural Networks and Deep Learning in Science - 09616 (Spring 24)

- Instructor: Dr Olexandr Isayev
- Tools Used: Python (OOP, NumPy, SciPy, Scikit-learn, PyTorch), Classification, Regression, Clustering, Boosting, Classic NN Architectures (ResNet, LeNet, AlexNet, VGG, CNN, RNN, etc)

Computational Modeling, Statistical Analysis and Machine Learning in Science - 09615 (Fall 22)

- Instructor: Dr Olexandr Isayev
- Tools Used: Python (NumPy, SciPy, Scikit-learn), Statistics (bias, variance, overfitting, regularization), Exploratory Data Analysis (elbow method, dimensionality reduction, outlier detection)

Awards

INSPIRE Fellowship Aug 2016 - May 2021

The Innovation in Science Pursuit for Inspired Research (INSPIRE) fellowhship awarded by the Department of Science and Technology (DST), Government of India to the 99th percentile students enrolled in public universities

NSF I-Corps Regional Aug 2025 - Sept 2025

This is a competitive NSF funded program designed to train academics to convert university spin-off products and technologies into start-ups through evidence based customer discovery and business model development plans.

Skills

Programming Languages:
Python, Bash, Git, HTML, CSS, JavaScript

Technical stack:
[*Data Science*] NumPy, SciPy, Scikit-learn, dimensionality reduction methods (PCA, tSNE, UMAP)
[*Machine Learning*] PyTorch for classification, regression and generation of audio/vision/molecular inputs
[*Cloud Computing*] Amazon Web Services (AWS), Google Cloud Platform (GCP), Kaggle, Weights and Biases (WandB)
[*Computational Chemistry*] RDKit, Density Functional Theory (DFT), Machine Learned Interatomic Potential (MLIP)
[*Webdev*] Nginx, Flask, Dashapp, Plotly, SQLite

Chemistry:
[*Synthetic*] Organic and inorganic synthesis, chromatographic purification techniques, air and moisture sensitive techniques (Schlenk line and glove box), distillation and crystallization (vapor diffusion and layering)
[*Analytical*] 1D and 2D NMR, cyclic voltammetry, open-circuit potential, UV-vis spectroscopy, Mössbauer, x-ray diffraction, elemental analysis, mass spectrometry (electron-spray ionization and gas chromatography), infrared spectroscopy